



THE KING'S SCHOOL

2008
Higher School Certificate Course
Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-10
- All questions are of equal value

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Total marks – 120
Attempt Questions 1-10
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.	Marks
(a) Differentiate xe^x	2
(b) $f(x) = x^3 - 3x^2 - 6x - 6$ Find $f'(-1)$	2
(c) Solve $(x - \sqrt{2})^2 = 4$	2
(d) Find a primitive of $\sec^2 2x$	2
(e) Find, correct to two decimal places, $\log_{12} 2008$	2
(f) Factorise $a^3 + 8b^3$	2

End of Question 1

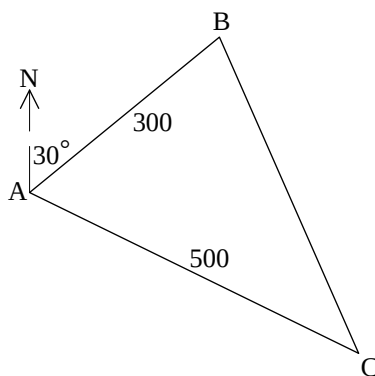
(a) Solve $\tan\theta + 1 = 0$ for $0 < \theta < 2\pi$ 2

(b) Find

(i) $\int \frac{x}{x^2 + 1} dx$ 2

(ii) $\int \frac{x^2 + 1}{x} dx$ 2

(c)



A ship sails from A for 300 km on a bearing of 030° to B. Another ship sails from A for 500 km on a bearing of 150° to C.

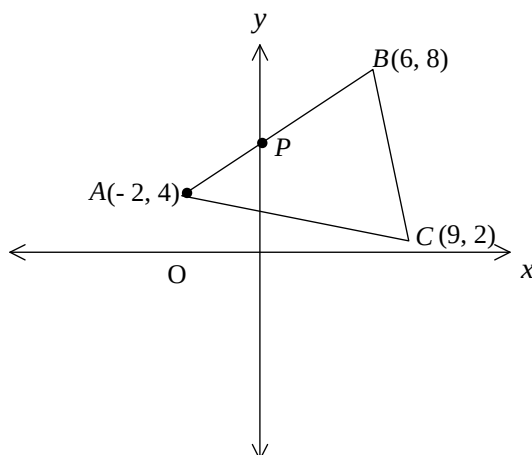
(i) Show that $\angle BAC = 120^\circ$. 1

(ii) Use the cosine rule to show that $BC = 700$ km. 2

(iii) Use the sine rule to find the bearing of C from B.
Give your answer correct to the nearest degree. 3

End of Question 2

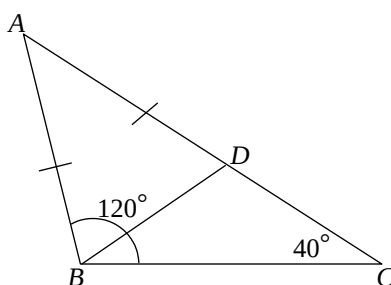
(a)



The vertices of $\triangle ABC$ are $(-2, 4)$, $(6, 8)$, $(9, 2)$, respectively. The line AB meets the y axis at P .

- | | |
|---|----------|
| (i) Find the gradient of AB . | 1 |
| (ii) Hence, or otherwise, find the y coordinate of P . | 2 |
| (iii) By finding the gradient of BC , or otherwise, show that $\angle ABC = 90^\circ$. | 2 |
| (iv) Find the area of $\triangle ABC$. | 2 |
| (v) Hence find the height of the triangle using BC as its base. | 1 |

(b)



In the diagram, $\angle BCA = 40^\circ$, $\angle ABC = 120^\circ$, $AB = AD$

- | | |
|--|----------|
| (i) Prove that $\angle ADB = 80^\circ$ | 2 |
| (ii) Deduce that $BD = CD$ | 2 |

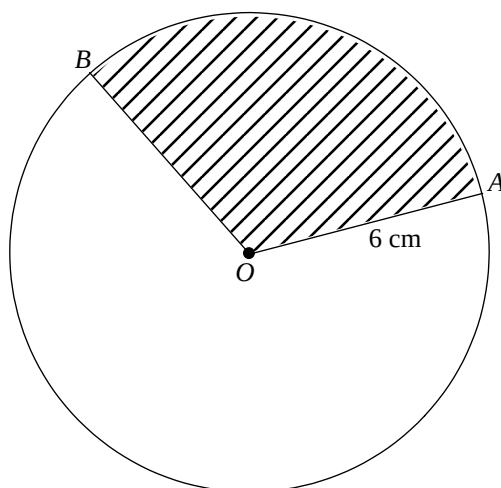
End of Question 3

- (a) The roots of $x^2 - 12x + 6 = 0$ are α, β .

Evaluate $\alpha + \beta + \frac{1}{\alpha} + \frac{1}{\beta}$

3

- (b)



The diagram shows a sector of a circle, centre O , and radius 6 cm. The area of the sector is $12\pi \text{ cm}^2$.

- (i) Find $\angle AOB$ **2**
- (ii) Find the length of the arc AB of the sector. **1**
- (c) A particular curve $y = f(x)$ has a stationary point $(0, 0)$. Also, $f''(x) = 2e^{2x} - 2$.
- (i) Show that at $(0, 0)$ there is a horizontal point of inflection. **2**
- (ii) Show that $f'(x) = e^{2x} - 2x - 1$ **2**
- (iii) Find $f(1)$ **2**

End of Question 4

(a) (i) Sketch the hyperbola $y = \frac{1}{x+1}$ showing any intercepts made on the axes. 2

(ii) The region bounded by the hyperbola $y = \frac{1}{x+1}$ and the x axis from $x = 0$ to $x = k > 0$ is revolved about the x axis.

Prove that the volume of the solid of revolution is less than π for all values of $k > 0$. 4

(b) Let $P(x) = -x^2 + (k+2)x - 1$

(i) Sketch $y = P(x)$ for $k = -4$ 2

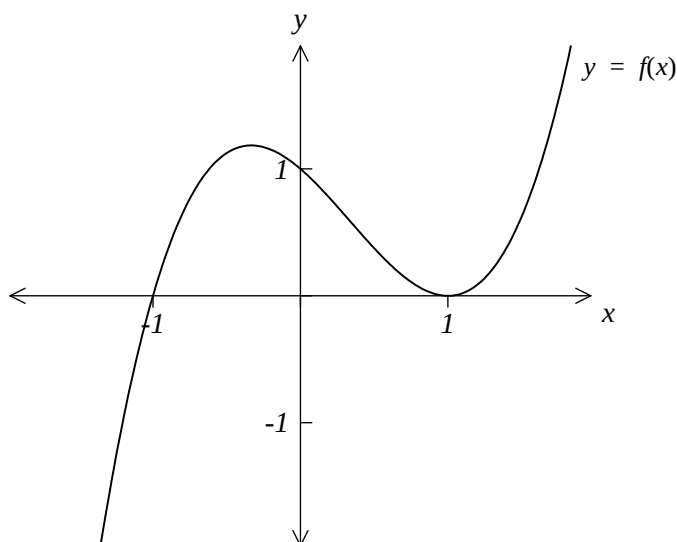
(ii) For what values of k does $P(x) = 0$ have real roots? 4

End of Question 5

- (a) Find the equation of the tangent to the curve $y = \sin(x^2 + 2x)$ at the point where $x = -2$.

4

(b)



The diagram shows the sketch of $y = f(x)$.

Sketch the graph of $y = f'(x)$.

2

- (c) $P(198, 998)$ is a point on the parabola $(x + 2)^2 = 4000(1008 - y)$. The directrix of the parabola is the line $y = 2008$.

Find the distance from P to the focus.

2

- (d) Simplify $\frac{a}{(a-b)(a-c)} - \frac{b}{(b-c)(a-b)} + \frac{c}{(a-c)(b-c)}$

2

- (e) Find the range of the function $y = \ln(2 + \sin x)$.

2

End of Question 6

- (a) Use Simpson's Rule with three function values to give a two decimal place

approximation to $\int_0^{\frac{\pi}{3}} 2\tan^3 x \, dx$.

3

- (b) (i) Show that $\frac{d}{dx}(\tan^2 x) = 2\tan x + 2\tan^3 x$

2

- (ii) Hence, or otherwise, find $\frac{d}{dx}(\sec^2 x)$

1

- (iii) Use (i) to show that $\int_0^{\frac{\pi}{3}} 2\tan^3 x \, dx = 3 - 2\ln 2$

3

- (c) $1 + 2x + 4x^2 + \dots$ is a geometric series.

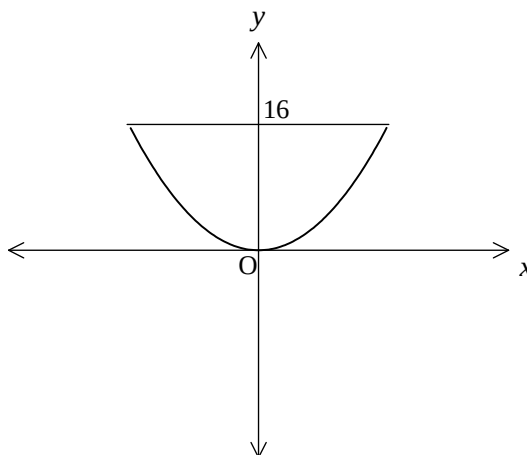
- (i) For what values of x does the limiting sum exist?

1

- (ii) Is it possible for the limiting sum to be 12? Give reasons.

2**End of Question 7**

(a)



The diagram shows the region enclosed between the curve $y = 16x^4$ and the line $y = 16$.

Find the area of the region.

3

- (b) (i) Albert deposited \$2 000 each year for 20 years into a fund paying 12% p.a. simple interest. Find the interest Albert made over the 20 years.

3

- (ii) Betty deposited \$2 000 each year for 20 years into a fund paying 7% p.a. compound interest. The interest was compounded annually.

Who made the better financial decision? Albert or Betty?

4

- (c) Solve $\ln x^2 - \ln x - 12 = 0$

2**End of Question 8**

- (a) A cylinder, open at one end, is to have a volume of $1728\pi \text{ cm}^3$.

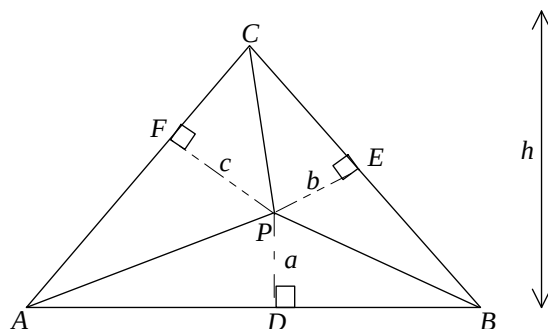
$$[V = \pi r^2 h, \text{ Curved Surface Area} = 2\pi rh]$$

Let the radius of the cylinder be r cm and the height h cm.

- (i) Show that the total surface area S is given by $S = \pi\left(r^2 + \frac{3456}{r}\right)$. 2
- (ii) Prove that the minimum surface area is $432\pi \text{ cm}^2$. 4
- (b) Sketch the curve $y = -\cos\left(\frac{x}{2}\right)$ for $-4\pi \leq x \leq 4\pi$. 2
- (c) (i) Solve $|x + 2| \leq 5$. 1
- (ii) Hence, or otherwise, solve $1 \leq |x + 2| \leq 5$. 3

End of Question 9

(a) (i)



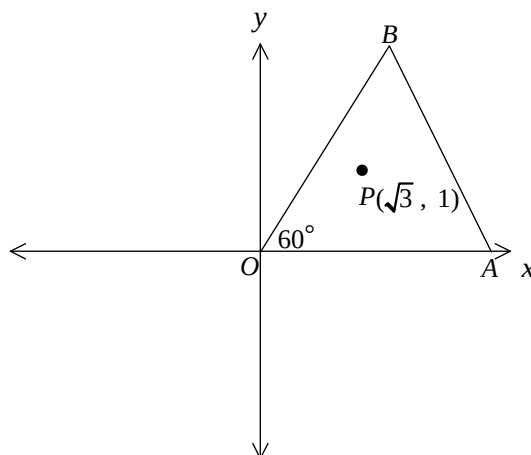
ΔABC is equilateral and P is any point interior to the triangle.

Perpendiculars PD , PE and PF are drawn to the sides AB , BC and CA , respectively.

Let $PD = a$, $PE = b$ and $PF = c$

If h is the altitude of the triangle, by considering areas of triangles, show that $h = a + b + c$.

2



ΔOAB is equilateral. O is the origin. A is on the x axis and B has the x coordinate 2.

$P(\sqrt{3}, 1)$ is a point interior to the triangle.

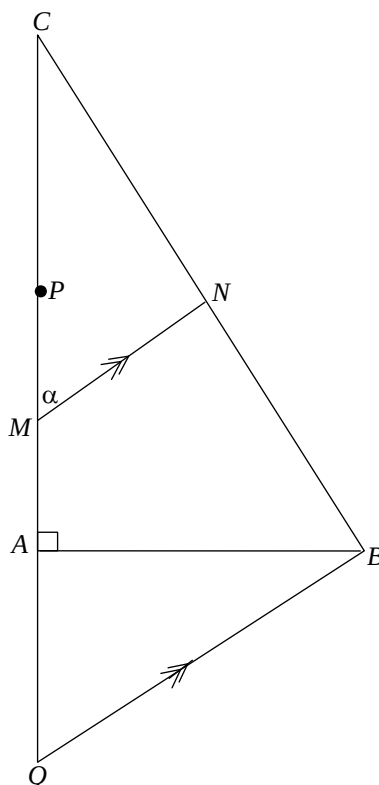
(ii) Show that the equation of OB is $y = \sqrt{3}x$. 2

(iii) Find the perpendicular distance from $P(\sqrt{3}, 1)$ to the line OB . 2

(iv) Hence, or otherwise, find the perpendicular distance from $P(\sqrt{3}, 1)$ to the line AB . 2

Question 10 continues on next page

(b)



The diagram shows $\triangle ABC$, right-angled at A . P is the point on CA so that $CP = AB$.

M is the mid-point of AP and N is the mid-point of BC .

Q is the point on CA produced so that QB is parallel to MN .

Let $\angle CMN = \alpha$.

Prove that α is a fixed value.

4

End of Examination

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Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note: $\ln x = \log_e x, \quad x > 0$



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Mathematics

Question	Algebra and Number	Geometry	Functions	Trigonometry	Differential Calculus	Integral Calculus	Total
1	(c), (e), (f) 6				(a), (b) 4	(d) 2	12
2				(a), (c) 8		(b) 4	12
3		(b) 4	(a) 8				12
4			(a), (b) 6			(c) 6	12
5			(a)(i), (b) 8			(a)(ii) 4	12
6	(d) 2		(c), (e) 4		(a), (b) 6		12
7	(c) 3				(d)(i)(ii) 3	(a), (b)(iii) 6	12
8	(b), (c) 9					(a) 3	12
9	(c) 4			(b) 2	(a) 6		12
10	(a) 8	(b) 4					12
Total	32	8	26	10	19	25	120

Question 1

$$(a) \quad x e^x + e^x$$

$$(b) \quad f'(x) = 3x^2 - 6x - 6$$

$$\therefore f'(-1) = 3 + 6 - 6 = 3$$

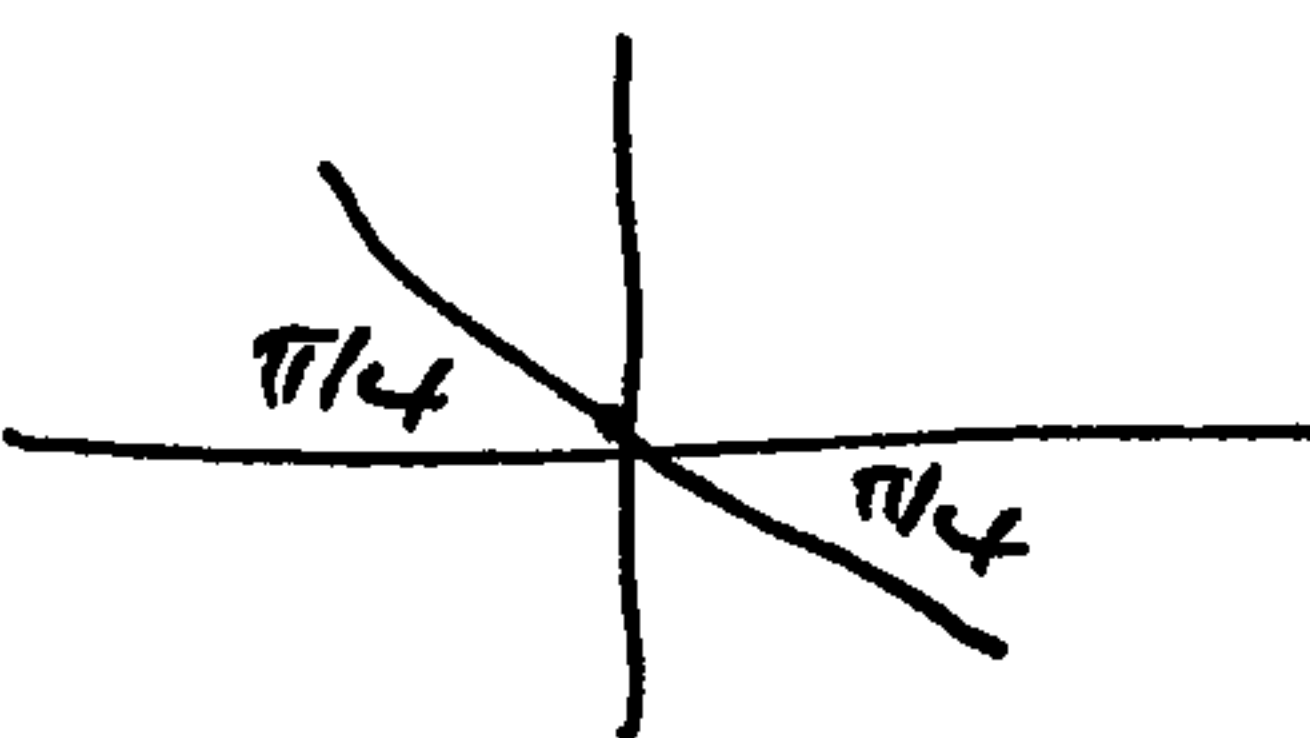
$$(c) \quad x - \sqrt{2} = \pm 2 \quad \therefore x = \sqrt{2} \pm 2$$

$$(d) \quad \frac{1}{2} \tan 2x$$

$$(e) \quad \log_{12} 2008 = \frac{\ln 2008}{\ln 12} = 3.06, 2 \text{ d.p.}$$

$$(f) \quad (a + 2b)(a^2 - 2ab + 4b^2)$$

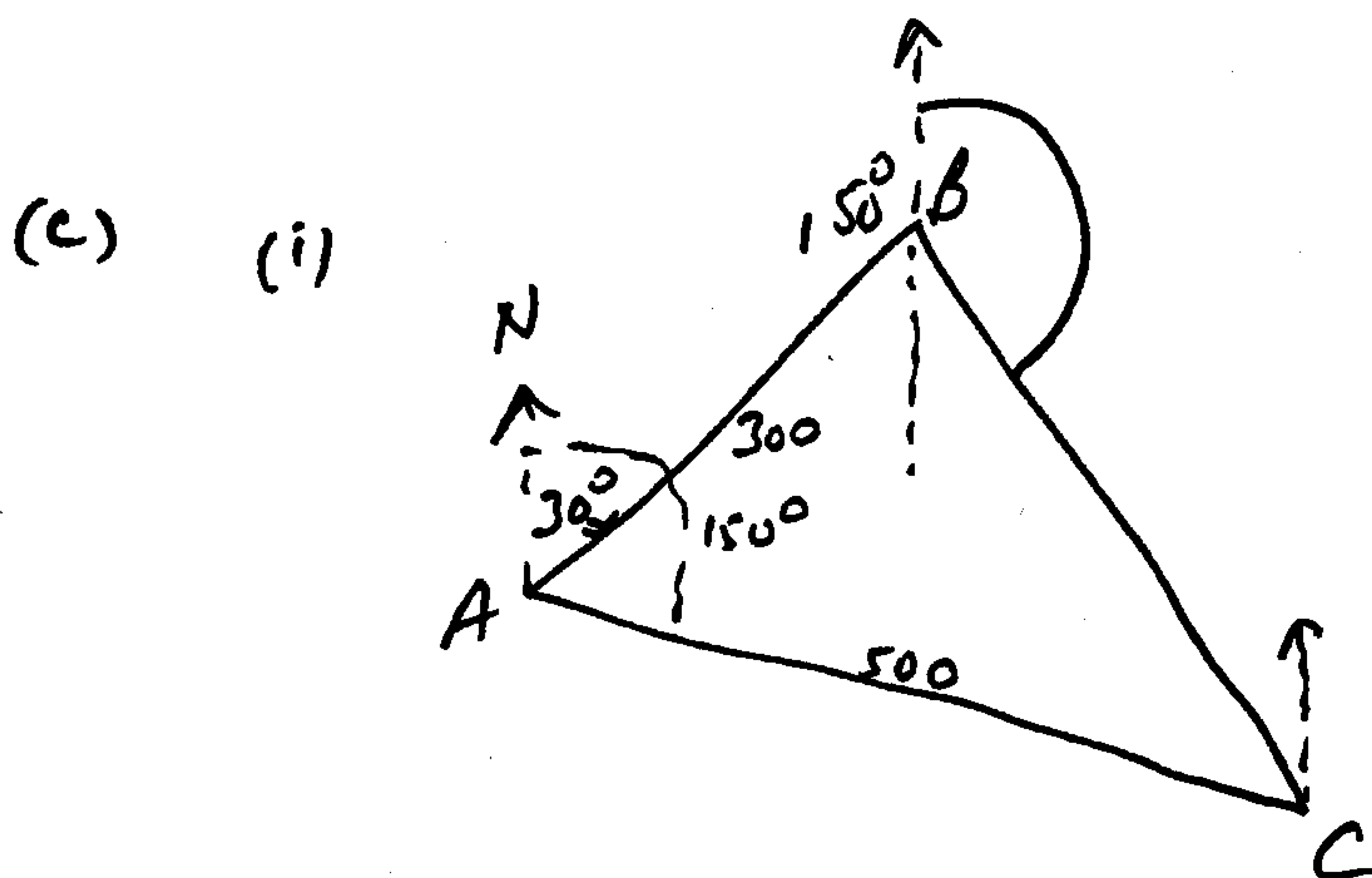
Question 2

(a) $\therefore \tan \theta = -1 \Rightarrow$ 

$\therefore \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$

(b) (i) $\frac{1}{2} \int \frac{2x}{x^2+1} dx = \frac{1}{2} \ln(x^2+1)$

(ii) $\int x + \frac{1}{x} dx = \frac{x^2}{2} + \ln x$



$$\angle BAC = \angle NAC - \angle NAB = 150^\circ - 30^\circ = 120^\circ$$

(ii) $BC^2 = 300^2 + 500^2 - 2 \times 300 \times 500 \cos 120^\circ$
 $= 490000$

$$\Rightarrow BC = \sqrt{490000} = 700$$

(iii) $\frac{\sin B}{500} = \frac{\sin 120^\circ}{700} \therefore \sin B = \frac{500 \sin 120^\circ}{700}$

$$\hat{B} = 38^\circ, \text{ nearest degree}$$

$\therefore \text{bearing of C from B} = 360^\circ - (150^\circ + 38^\circ)$
 $= 172^\circ$

Question 3

(a) (i) $\text{grad } AB = \frac{8-4}{6--2} = \frac{4}{8} = \frac{1}{2}$

(ii) take $P(0, y) \Rightarrow \frac{y-4}{0--2} = \frac{1}{2}$

$$\therefore y-4=1, \quad y=5$$

(iii) $\text{grad } BC = \frac{6}{-3} = -2$

* since $\text{grad } AB \times \text{grad } BC = -1$, then $AB \perp BC$

$$\Rightarrow \angle ABC = 90^\circ$$

(iv) $AB = \sqrt{8^2 + 4^2} = \sqrt{80}$

$$BC = \sqrt{3^2 + 6^2} = \sqrt{45}$$

$$\therefore \text{Area } \triangle ABC = \frac{1}{2} \cdot \sqrt{80} \cdot \sqrt{45} = 30$$

(v) $\therefore 30 = \frac{1}{2} \cdot \sqrt{45} \cdot h \Rightarrow h = \frac{60}{\sqrt{45}}$ will do $\left[\begin{array}{l} \text{or } \frac{20}{\sqrt{5}} \\ \text{or } 4\sqrt{5} \end{array} \right]$

(b) (i) $\angle BAC = 180^\circ - (120^\circ + 40^\circ)$, $\angle \text{sum } \triangle ABC$
 $= 20^\circ$

In isosceles $\triangle ABD$, $\angle B = \angle D = \frac{1}{2} (160^\circ) = 80^\circ$,

$$\angle \text{sum } \triangle ABD$$

$$\therefore \angle ADB = 80^\circ$$

(ii) $\angle DBC + \angle DCB = \angle ADB$, ext $\angle$ theorem in $\triangle BDC$

$$\Rightarrow \angle DBC = 80^\circ - 40^\circ = 40^\circ$$

$\therefore \triangle BDC$ is isosceles, base angles equal

$$\therefore BD = CD$$

Question 4

(a) $\alpha + \beta = 12, \alpha\beta = 6$

$$\therefore \alpha + \beta + \frac{1}{\alpha} + \frac{1}{\beta} = \alpha + \beta + \frac{\alpha + \beta}{\alpha\beta} = 12 + 2 = 14$$

(b) (i) Let $\angle AOB = \theta$

$$\text{Then } \frac{1}{2} \cdot 6^2 \cdot \theta = 12\pi \Rightarrow \theta = \frac{12\pi}{18} = \frac{2\pi}{3}$$

(ii) $AB = 6 \cdot \frac{2\pi}{3} \text{ cm} = 4\pi \text{ cm}$

(c) (i) $f''(0) = 2 - 2 = 0$

$$f''(-1) = 2e^{-2} - 2 < 0 \Rightarrow \text{change in concavity}$$

$$f''(1) = 2e^2 - 2 > 0$$

$\therefore (0, 0)$ is a horiz. pt. of inflection since it's a stat. pt

(ii) $f'(x) = 2 \frac{e^{2x}}{2} - 2x + c$

$$\therefore f'(0) = 1 - 0 + c = 0, c = -1$$

$$\therefore f'(x) = e^{2x} - 2x - 1$$

(iii) $f(x) = \frac{e^{2x}}{2} - x^2 - x + c$

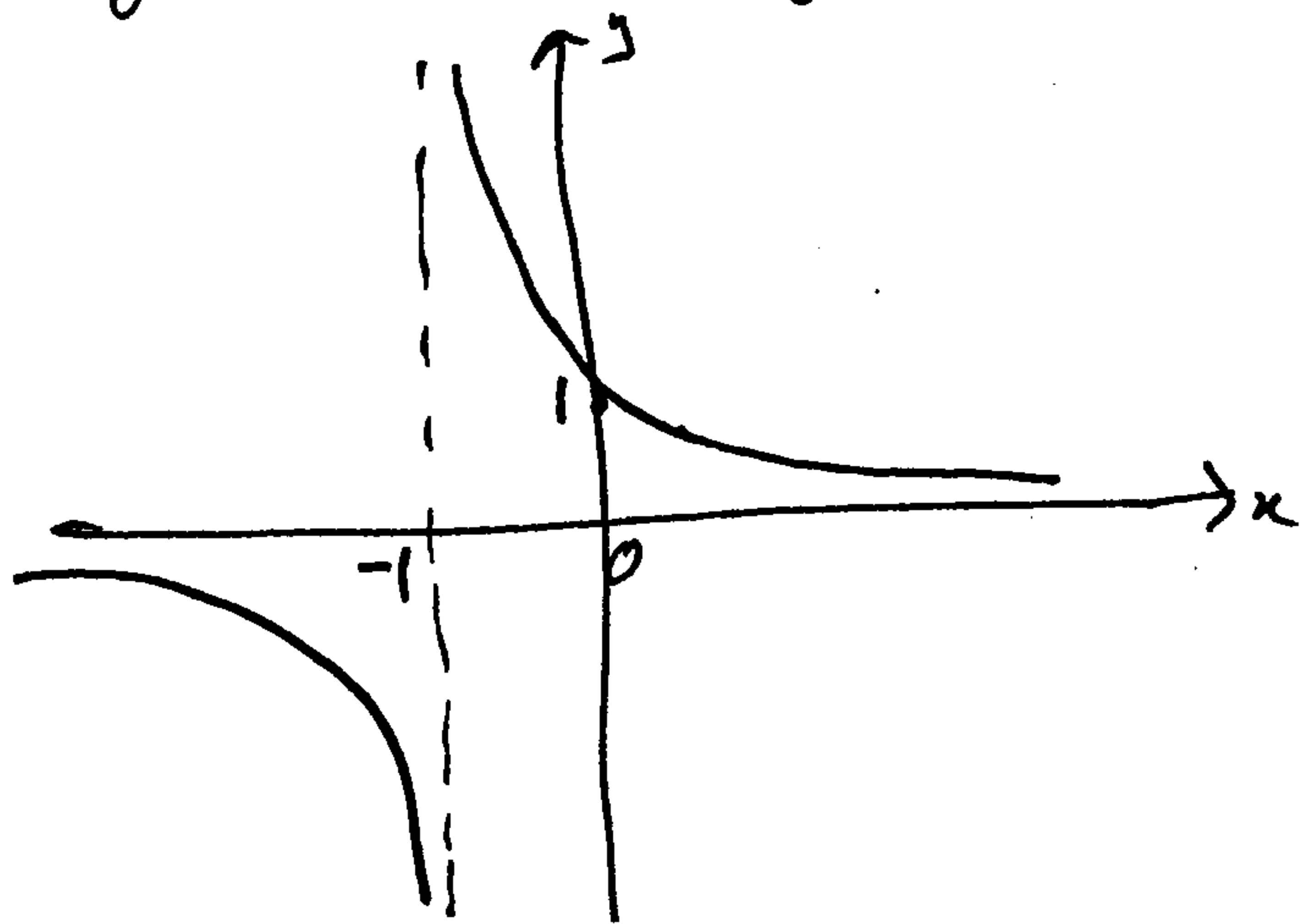
$$\therefore f(0) = \frac{1}{2} - 0 + c = 0, c = -\frac{1}{2}$$

$$\therefore f(x) = \frac{e^{2x}}{2} - x^2 - x - \frac{1}{2}$$

$$\therefore f(1) = \frac{e^2}{2} - 1 - 1 - \frac{1}{2} = \frac{e^2 - 5}{2}$$

Question 5

(a) (i) $x \neq -1, y \neq 0$; $x=0, y=1$



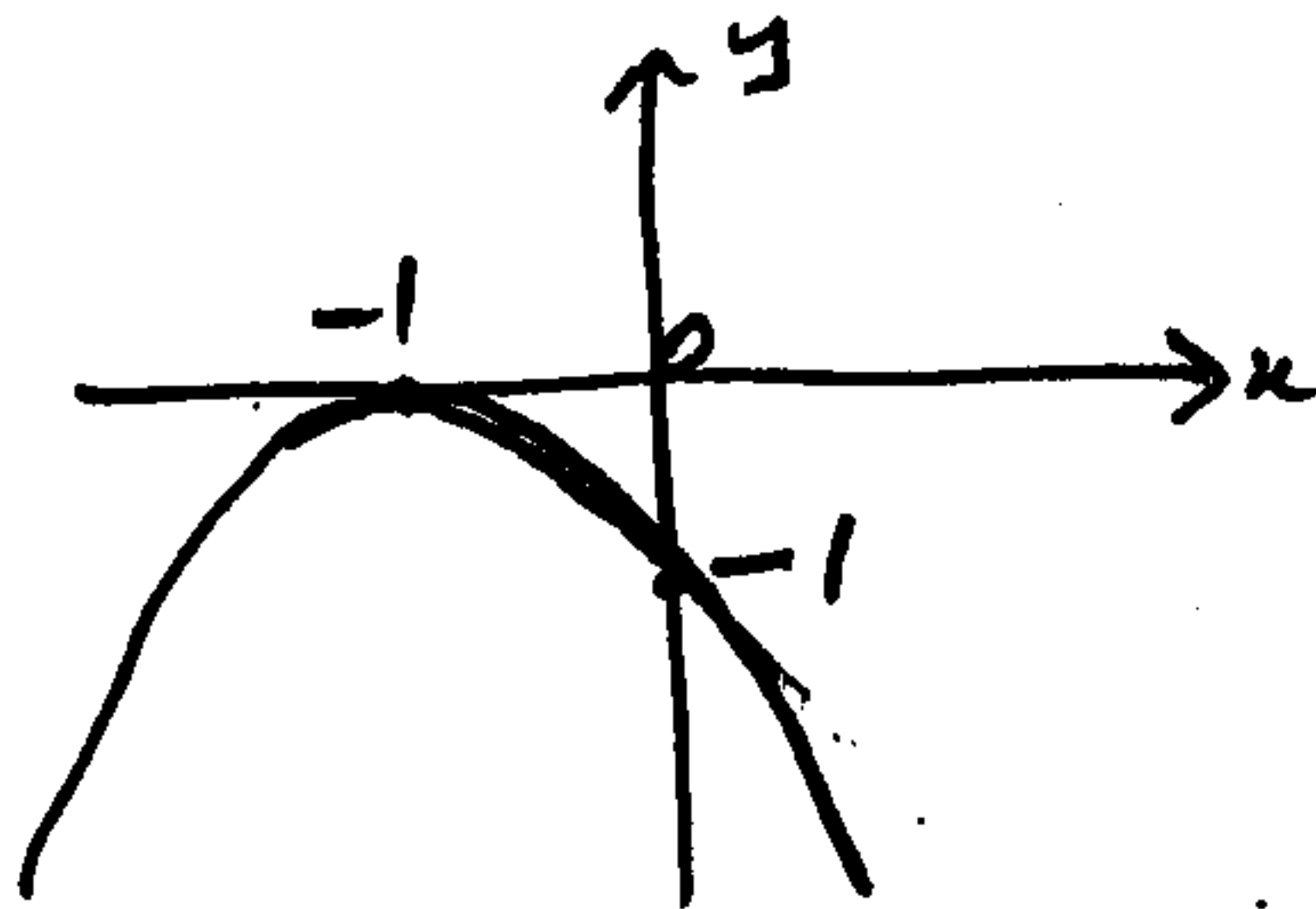
(ii) $V = \pi \int_0^k \left(\frac{1}{x+1} \right)^2 dx = \pi \int_0^k (x+1)^{-2} dx$

$$= \pi \left[\frac{(x+1)^{-1}}{-1} \right]_0^k$$

$$= \pi \left[-\frac{1}{x+1} \right]_0^k = \pi \left(-\frac{1}{k+1} + 1 \right)$$

$$= \pi \left(1 - \frac{1}{k+1} \right) < \pi \text{ since } \frac{1}{k+1} > 0$$

(b) (i) $y = -x^2 - 2x - 1 = -(x^2 + 2x + 1) = -(x+1)^2$



(ii) $\Delta = (k+2)^2 - 4(-1)(-1) = (k+2)^2 - 4$

$$= k^2 + 4k$$

$$= k(k+4) \geq 0 \text{ for real roots}$$

$$\therefore k \leq -4 \text{ or } k \geq 0$$

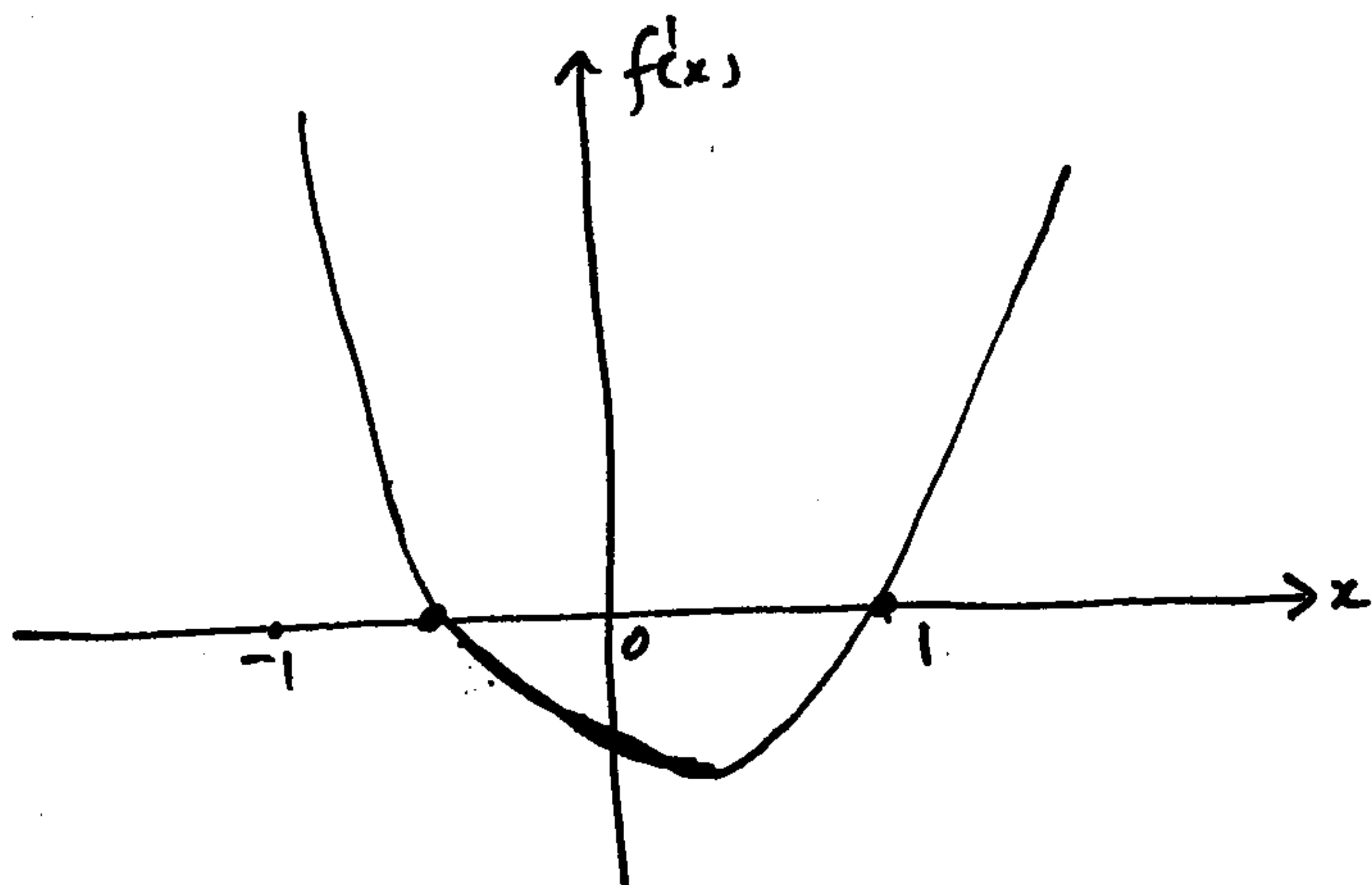
Question 6

(a) $\frac{dy}{dx} = (2x+2) \cos(x^2+2x)$

For $x = -2$, $y = \sin 0 = 0$, $\frac{dy}{dx} = -2 \cos 0 = -2$

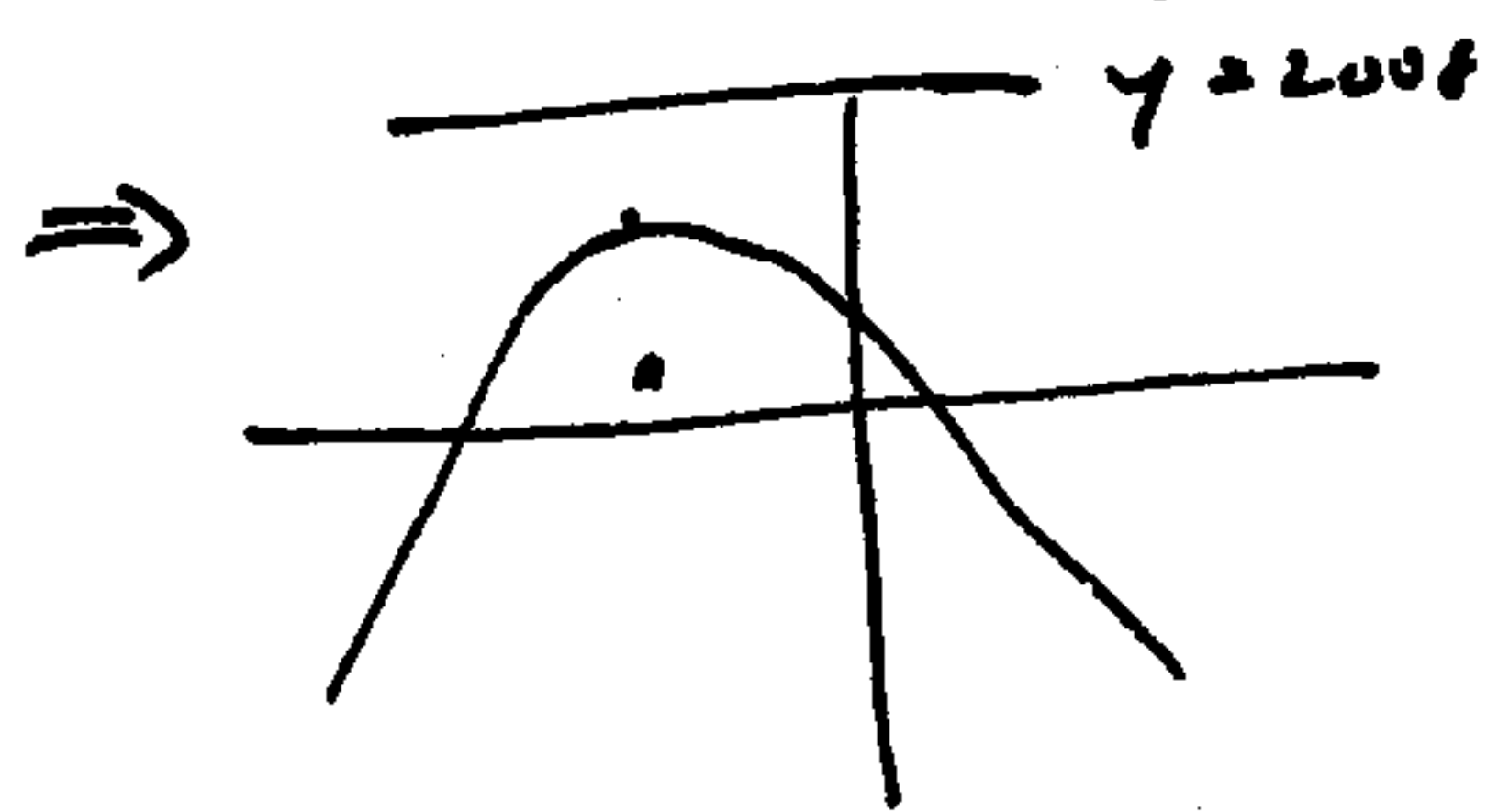
$\therefore$ tangent is $y = -2(x+2)$

(b)



(c) P to the focus = P to the directrix
 $= 2008 - 998 = 1010$

OR (sadly) $V = (-2, 1008)$; $a = 1000 \Rightarrow S = (-2, 0)$



$\therefore PS = \sqrt{200^2 + 998^2} = 1010$

(d)
$$\frac{a(b-c) - b(a-c) + c(a-b)}{(a-b)(a-c)(b-c)} = \frac{ab - ac - ab + bc + ac - bc}{(a-b)(a-c)(b-c)} = 0$$

(e) Since $-1 \leq \sin x \leq 1$ and $\ln x$ is an increasing function,

range y : $\ln(2-1) \leq y \leq \ln(2+1)$

$\therefore 0 \leq y \leq \ln 3$

Question 7

$$(a) \int_0^{\frac{\pi}{3}} 2 \tan^3 x \, dx \approx \frac{1}{6} \cdot \frac{\pi}{3} \left(0 + 2 \tan^3 \frac{\pi}{3} + 8 \tan^3 \frac{\pi}{6} \right) \\ = 2.08, \quad 2 \text{ d.p.}$$

$$(b) (i) \frac{d(\tan^2 x)}{dx} = 2 \tan x \sec^2 x \\ = 2 \tan x (1 + \tan^2 x) = 2 \tan x + 2 \tan^3 x$$

$$(ii) \frac{d(\sec^2 x)}{dx} = \frac{d(1 + \tan^2 x)}{dx} = 2 \tan x + 2 \tan^3 x \quad \text{or equivalent}$$

$$(iii) 2 \tan^3 x = \frac{d(\tan^2 x)}{dx} - 2 \tan x$$

$$\therefore \int_0^{\frac{\pi}{3}} 2 \tan^3 x \, dx = \left[\tan^2 x \right]_0^{\frac{\pi}{3}} - 2 \int_0^{\frac{\pi}{3}} \tan x \, dx \\ = (\sqrt{3})^2 - 0 + 2 \int_0^{\frac{\pi}{3}} \frac{-\sin x}{\cos x} \, dx \\ = 3 + 2 \left[\ln \cos x \right]_0^{\frac{\pi}{3}} \\ = 3 + 2 \left(\ln \frac{1}{2} - \ln 1 \right) \\ = 3 + 2 \ln 2^{-1} = 3 - 2 \ln 2$$

$$(c) (i) S_{\infty} \text{ exists for } -1 < 2x < 1 \quad \text{i.e. } -\frac{1}{2} < x < \frac{1}{2}$$

$$(ii) \text{ Put } S_{\infty} = \frac{1}{1-2x} = 12$$

$$\therefore 1-2x = \frac{1}{12}$$

$$2x = \frac{11}{12}$$

$$x = \frac{11}{24} < \frac{1}{2}$$

$\therefore$ from (i), it's possible

Question 8

$$(a) \quad A = 2 \int_0^{16} x \, dy, \quad x \geq 0 \quad : \quad x^4 = \frac{y}{16}$$
$$\text{ie. } x = \frac{y^{\frac{1}{4}}}{2}$$

$$\therefore A = \int_0^{16} y^{\frac{1}{4}} \, dy = \frac{4}{5} \left[y^{\frac{5}{4}} \right]_0^{16} = \frac{4}{5} \cdot 2^5 = \frac{128}{5}$$

$$(b) \quad (i) \text{ Albert made } \$2000(0.12)(20) + 2000(0.12)(19) + \dots + 2000(0.12)$$

$$= \$240(1+2+\dots+19+20)$$

$$= \$240 \cdot \frac{20 \cdot 21}{2} = \$50400 \text{ interest}$$

(ii) Betty's total over the 20 years

$$= \$2000(1.07)^{20} + 2000(1.07)^{19} + \dots + 2000(1.07)^2 + 2000(1.07)$$

$$= \$2000(1.07) [1 + 1.07 + 1.07^2 + \dots + 1.07^{19}]$$

$$= \$2000(1.07) \frac{1.07^{20} - 1}{0.07} = \$87730, \text{ nearest dollar}$$

$$\therefore \text{Interest made} = \$87730 - \$2000 \times 20 = \$47730$$

$\therefore$ Albert's decision was better

$$(c) \quad \therefore 2 \ln x - \ln x - 12 = 0$$

$$\ln x = 12$$

$$x = e^{12}$$

Question 9

(2) (i) $S = 2\pi rh + \pi r^2$: $\pi r^2 h = 1728\pi$

$$h = \frac{1728}{r^2}$$

$$\therefore S = 2\pi r \cdot \frac{1728}{r^2} + \pi r^2 = \pi \left(r^2 + \frac{3456}{r} \right)$$

(ii) $S = \pi (r^2 + 3456 r^{-1})$

$$\frac{dS}{dr} = \pi (2r - 3456 r^{-2})$$

$$\frac{d^2S}{dr^2} = \pi (2 + 6912 r^{-3})$$

$$\frac{dS}{dr} = 0 \Rightarrow r - \frac{1728}{r^2} = 0 \quad \text{or } r^3 = 1728$$

$$\Rightarrow r = \sqrt[3]{1728} = 12$$

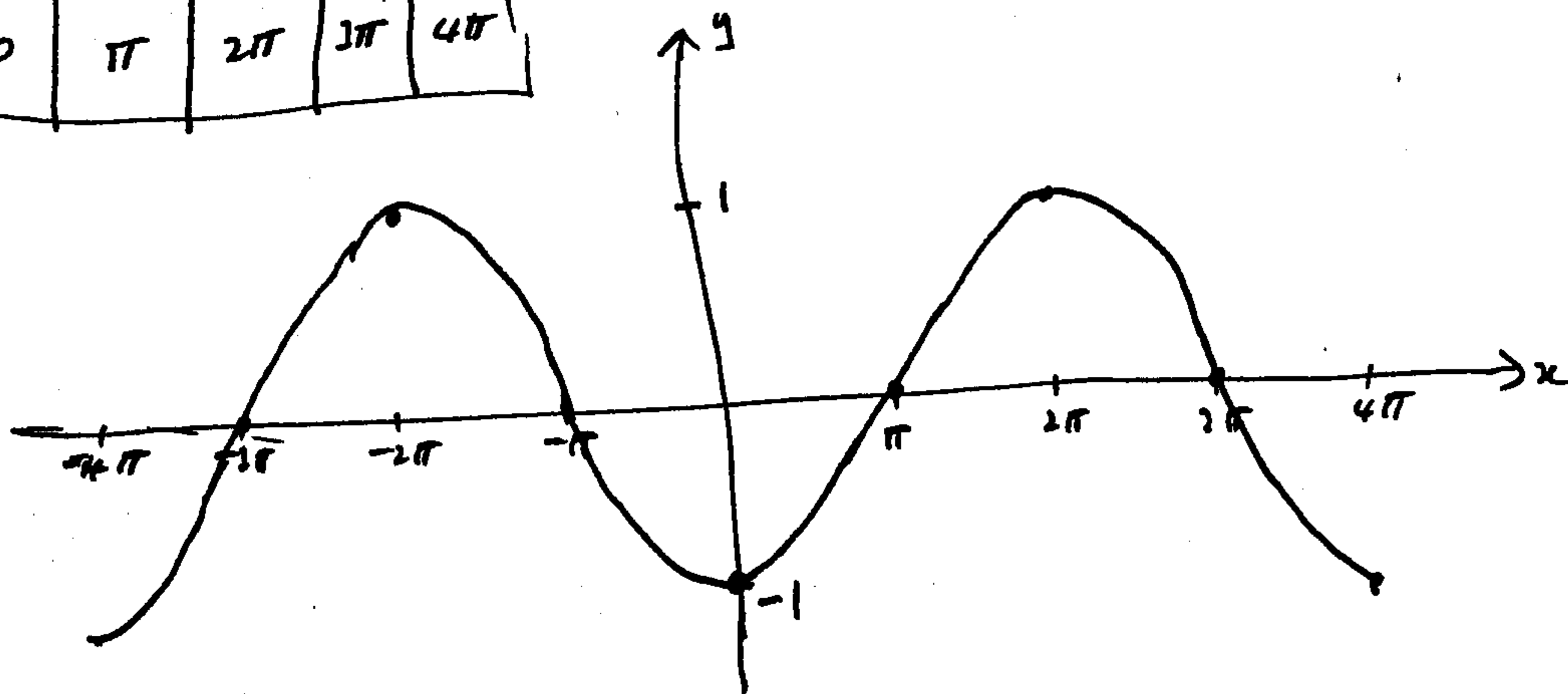
For $r=12$, $\frac{d^2S}{dr^2} > 0$ (indeed $\frac{d^2S}{dr^2} > 0$ for all $r > 0$)

$\Rightarrow$ minimum S when $r=12$

$$\therefore \min S = \pi \left(12^2 + \frac{3456}{12} \right) \text{ cm}^2 = 432\pi \text{ cm}^2$$

(4)

$\frac{x}{2}$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
y	-1	0	1	0	-1
x	0	π	2π	3π	4π



(C) (i)

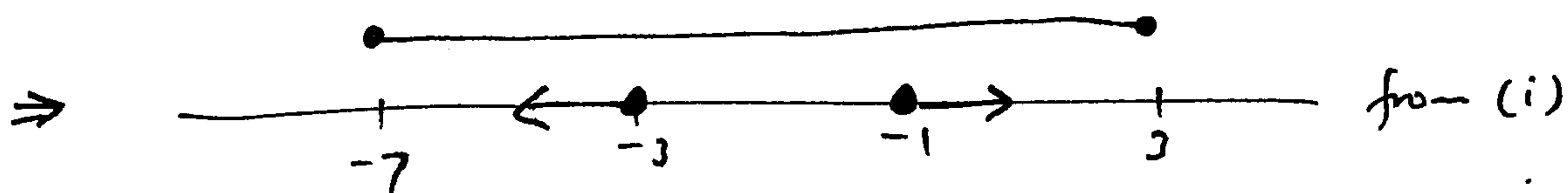
$$-5 \leq x+2 \leq 5$$

$$\therefore -7 \leq x \leq 3$$

(ii) For $|x+2| \geq 1$

we have $-1 \geq x+2 \geq 1$

$$-3 \geq x \geq -1 \quad \text{i.e. } x \leq -3 \text{ or } x \geq -1$$



$$\therefore -7 \leq x \leq -3 \quad \text{or} \quad -1 \leq x \leq 3$$

OR

$$1 \leq |x+2| \leq 5$$

$$\Rightarrow 1 \leq x+2 \leq 5$$

$$\text{or } -5 \leq x+2 \leq -1$$

$$\therefore -1 \leq x \leq 3$$

$$\text{or } -7 \leq x \leq -3$$

Question 10

$$(a) (i) \therefore \frac{1}{2} \cdot AB \cdot h = \frac{1}{2} \cdot AB \cdot a + \frac{1}{2} \cdot BC \cdot b + \frac{1}{2} \cdot AC \cdot c,$$

where $AB = BC = AC$, Δ equilateral

$$\therefore h = a + b + c$$

$$(ii) OB \text{ is } y - 0 = \tan 60^\circ (x - 0)$$

$$\text{i.e. } y = \sqrt{3}x$$

(iii) $\perp$ distance from $P(\sqrt{3}, 1)$ to $\sqrt{3}x - y = 0$ is

$$\frac{\sqrt{3} \cdot \sqrt{3} - 1}{\sqrt{\sqrt{3}^2 + 1^2}} = \frac{3-1}{\sqrt{4}} = 1$$

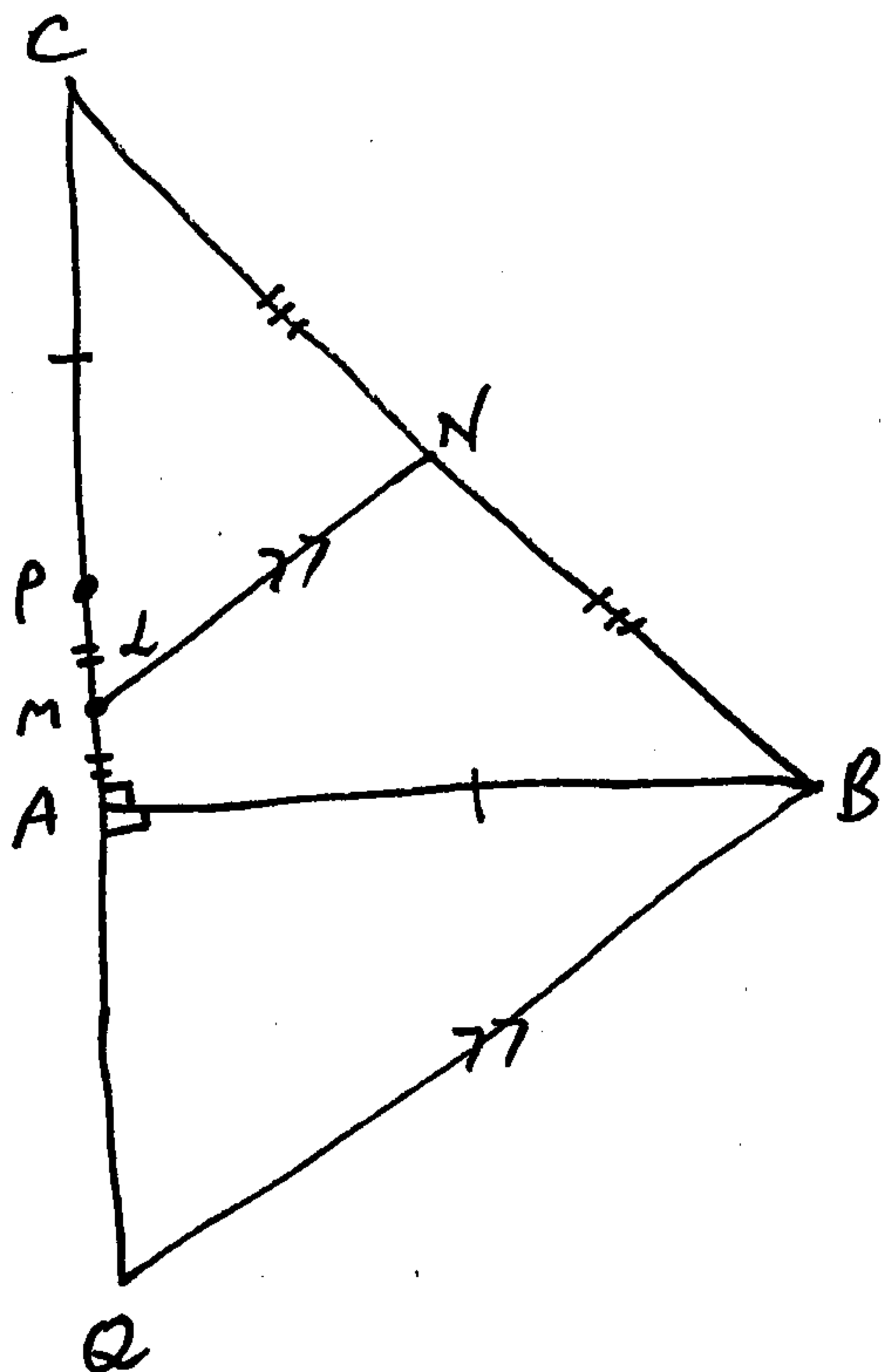
(iv) Using notation in (i), $c = 1$ from (iii)

$$a = 1 \text{ since } P = (\sqrt{3}, 1)$$

$$\text{at } B, x = 2 \therefore y = 2\sqrt{3} = h$$

$$\therefore \text{distance from } P \text{ to } AB = 2\sqrt{3} - 1 - 1 = 2\sqrt{3} - 2$$

(b)



$\angle BQA = \angle NMC = d$, corresponding $\angle$ s in $\parallel$ lines

$$\frac{BN}{NC} = \frac{QM}{MC} = 1:1, \text{ ratio intercept theorem in } \parallel \text{ lines}$$

$$\therefore QM = MC$$

But $AM = MP$, data

$$\therefore QA = PC = AB, \text{ data}$$

$\therefore \Delta ABC$ is right-angled and isosceles.

$$\Rightarrow \angle Q = \angle B = 45^\circ, \text{ base angles equal}$$

$$\therefore d \text{ is the fixed value } 45^\circ$$